CSDA5110
Exercise 2
Week 6 Fall 1 2024

I.

Sociology of Sport Journal (spring 1992) published research on Proposition 48, the NCAA regulation that requires a minimum of 700 on the SAT for eligibility for college athletics. Critics have claimed that the SAT is biased against black student athletes and is not a valid predictor of academic success. To investigate the validity of the SAT as a predictor of academic success, high school GPA and study time were also considered as potential predictors of college GPA.

College GPA = y
High school GPA = x1
Hours of studying in a season = x2
SAT scores = x3

The study conducted separate regression analyses for black and white student athletes. Some of the results are shown below.

	Parameter Estimates	**Standard Error**
Blacks		
Intercept	2.245	.09
High School GPA	-.040	.01
Study Hours	.14	.10
SAT score	.001	.0025
Whites		
Intercept	1.970	.16
High school GPA	-.089	.01
Study Hours	.20	.23
SAT Score	.001	.0002

a) Write the model relating the mean value of the college GPA as linear function of the three predictor variables. Explain the meaning of each of the parameter estimates in your model.

b) Note that the estimate of the regressor parameter for SAT score for both models is .001. Interpret the estimate.

c) The reported p-value for testing $H_0 : \beta_1 = 0$ is .734 for the black athletes' model and .000 for the white athletes' model. Interpret these values.

II.

A multiple regression model of the following form is fitted to a data set.

$$Y_i = \beta_0 + \beta_1 \cdot x_{i,1} + \beta_2 \cdot x_{i,2} + \beta_3 \cdot x_{i,3} + \beta_4 \cdot x_{i,4} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.}$$

The model is fitted using the software R and the following summary output is obtained.

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Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)      ???     0.1960   8.438 3.57e-13
x1            5.3036     2.5316     ??? 0.038834
x2            4.0336     2.4796   1.627 0.107111
x3           -9.3153     2.4657  -3.778 0.000276
x4            0.5884     2.2852   0.257 0.797373

Residual standard error: 1.892 on 95 degrees of freedom
Multiple R-squared: 0.1948,Adjusted R-squared: ???
F-statistic: 5.745 on 4 and 95 DF,  p-value: 0.0003483
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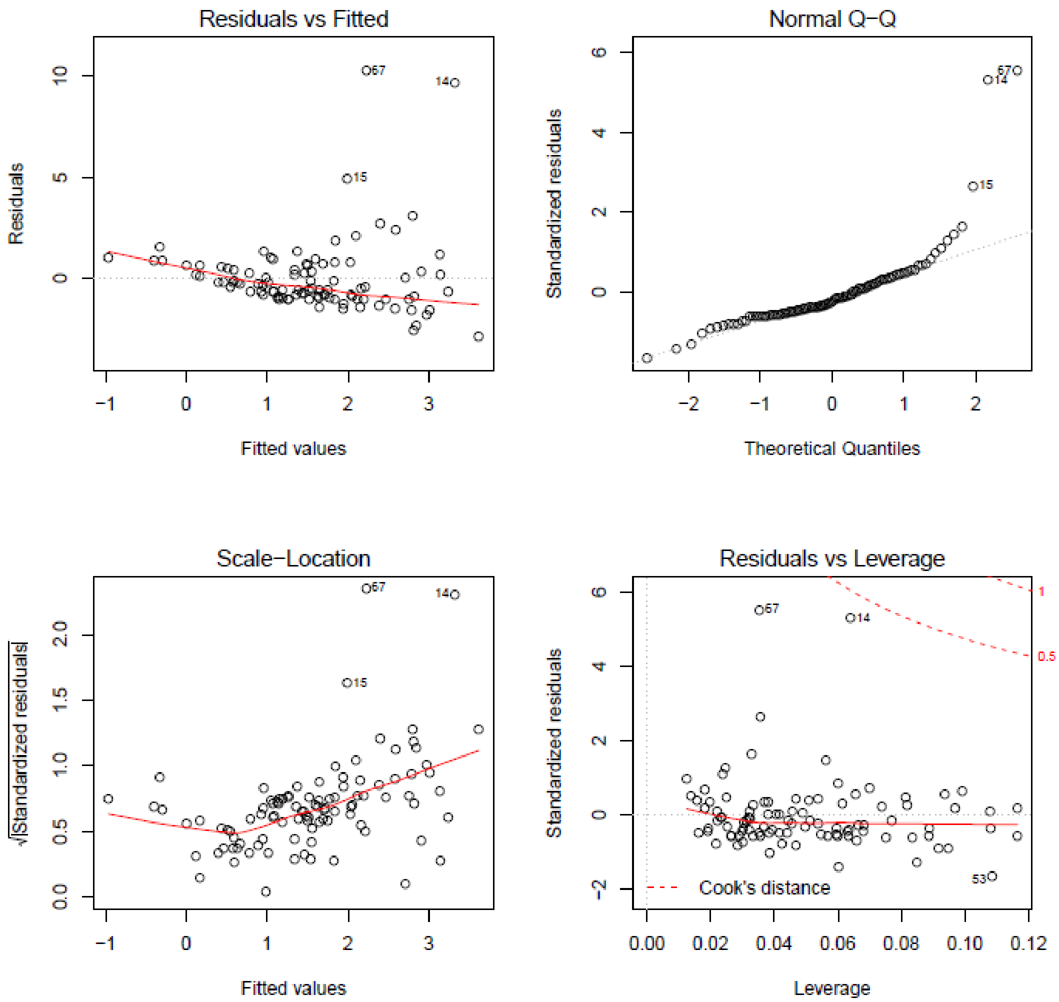
1) What is the value of the t-statistics of $\hat{\beta}_1$?
   a) 0.099 b) 13.43 c) 2.095 d) 0.015
2) How many observations are in the data set?
   a) 100 b) 99 c) 96 d) 95
3) Has the null hypothesis $H_0: \beta_3 = 0$ to be rejected on a 5% level?
   a) Yes b) No c) No answer possible.
4) What is the estimate of the intercept $\hat{\beta}_0$?
   a) 1.654 b) 0.324 c) 43.051 d) 1.591
5) What is the estimate of $Var(\epsilon_i)$.
   a) 1.892 b) 3.579 c) 1.375 d) 9.46
6) Which of the following intervals is a two-sided 95% confidence interval for β_3?
   a) $-9.315 \pm 1.99 \cdot 0.00028$ b) $-9.315 \pm 1.99 \cdot \frac{2.466}{\sqrt{95}}$
   c) $-9.315 \pm 1.99 \cdot \frac{0.00028}{\sqrt{95}}$ d) $-9.315 \pm 1.99 \cdot 2.466$
7) Have a look at the residual plots. Are the model assumptions on the ϵ_i fullfilled and if not, what is the main problem?
   a) Yes.
   b) No, since leverage points exist.
   c) No, since the assumption of constant variance of the ε_i is violated.
   d) No, since the ε_i are dependent.



Residuals vs Fitted
Residuals
Fitted values
Normal Q−Q
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage